

Types of Scientific Research

Interventional Designs

In an interventional design, scientists directly intervene with subjects and look at the impacts of the intervention.

True Experimental Design: RCTs

True experimental design consists of randomized controlled trials (RCTs), which are the "gold standard" of research due to their degree of rigor and control.

Parallel Design

Researchers track two or more independent groups in parallel with each other.



Factorial Design

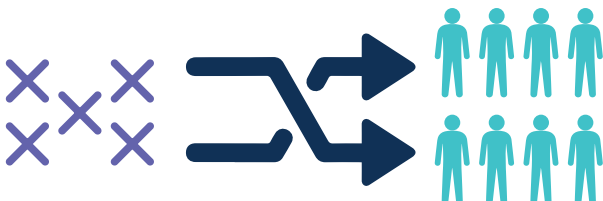
Researchers test the impact of two or more variables on one group simultaneously, rather than isolating a single variable.



Crossover Design

The same group of subjects are assigned to both the treatment and control groups. Each subject acts as their own control.

A strength of crossover designs is that they reduce the impact of the variability between subjects (e.g., the differences in how each person responds to a particular diet).

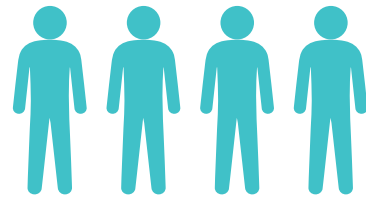


Quasi-experimental Designs

Quasi-experimental designs are similar to RCTs, but subjects are not randomly assigned to groups.

Non-randomized Control Group

Subjects have not been randomly assigned to that group. This carries more potential for bias than an RCT because the control group may not be equivalent to the treatment group.



Time-Series

Reserachers study a set of data points that have been collected in time order.

One example is the study of climate change, where average annual air or ocean temperatures are collected in sequence over several years.

